

Adaptive Capillary Skin (ACS) - Supplementary Information

Computational parameters, raw-data dictionary, physical build specification and
preregistered falsification protocol

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**SUPPLEMENT TO A COMPUTATIONAL DESIGN STUDY - NO PHYSICAL
EXPERIMENTS HAVE YET BEEN PERFORMED**

All numerical results in this document are generated by explicit computational models and assumptions. They must not be represented as measurements, proof of physical feasibility, or clinical/environmental performance. The document is designed to make the concept falsifiable before fabrication.

S1. Computational data dictionary

Table S1. Main data fields.

Column	Unit	Meaning
time_s	s	Simulation time; nominal damage begins at 300 s
Qin_m3_s	m^3 s^-1	Total modeled inlet flow from prescribed-pressure boundary
Qout_m3_s	m^3 s^-1	Useful modeled outlet flow to zero-pressure boundary
Qleak_m3_s	m^3 s^-1	Modeled flow lost through central damage conductance
P_damage_Pa	Pa	Pressure at damaged central node
sensor_voltage_V	V	Idealized electrokinetic surrogate including assumed Gaussian readout noise in nominal trace
sensor_deltaP_Pa	Pa	Pressure drop used by idealized sensing surrogate
leak_conductance_m3_Pa_s	m^3 Pa^-1 s^-1	Time-dependent leak conductance
hydrogel_swelling_fraction	dimensionless	Phenomenological 0-1 swelling-state proxy
pass_all	boolean	True only if every preregistered computational gate passes

S2. Core equations and assumptions

$$G = \pi r^4 / (8 \mu L) \quad (S1)$$

$$Q_{ij} = G_{ij}(P_i - P_j) \quad (S2)$$

$$\Delta P_{cap} = 2 \gamma \cos(\theta) / r \quad (S3)$$

$$G_I(t) = G_{min} + (G_0 - G_{min}) \exp[-(t - t_d - t_{delay}) / \tau] \quad (S4)$$

$$\Delta V = (\epsilon \zeta / (\mu \sigma)) \Delta P \quad (S5)$$

Assumption hierarchy: Eq. S1-S2 are standard laminar hydraulic relations; Eq. S3 is a scale estimate for capillary driving pressure; Eq. S4 is the principal unvalidated material-response hypothesis; Eq. S5 is an idealized electrokinetic surrogate whose practical value depends on electrode and pore physics not resolved here.

S3. Falsification parameter sweep

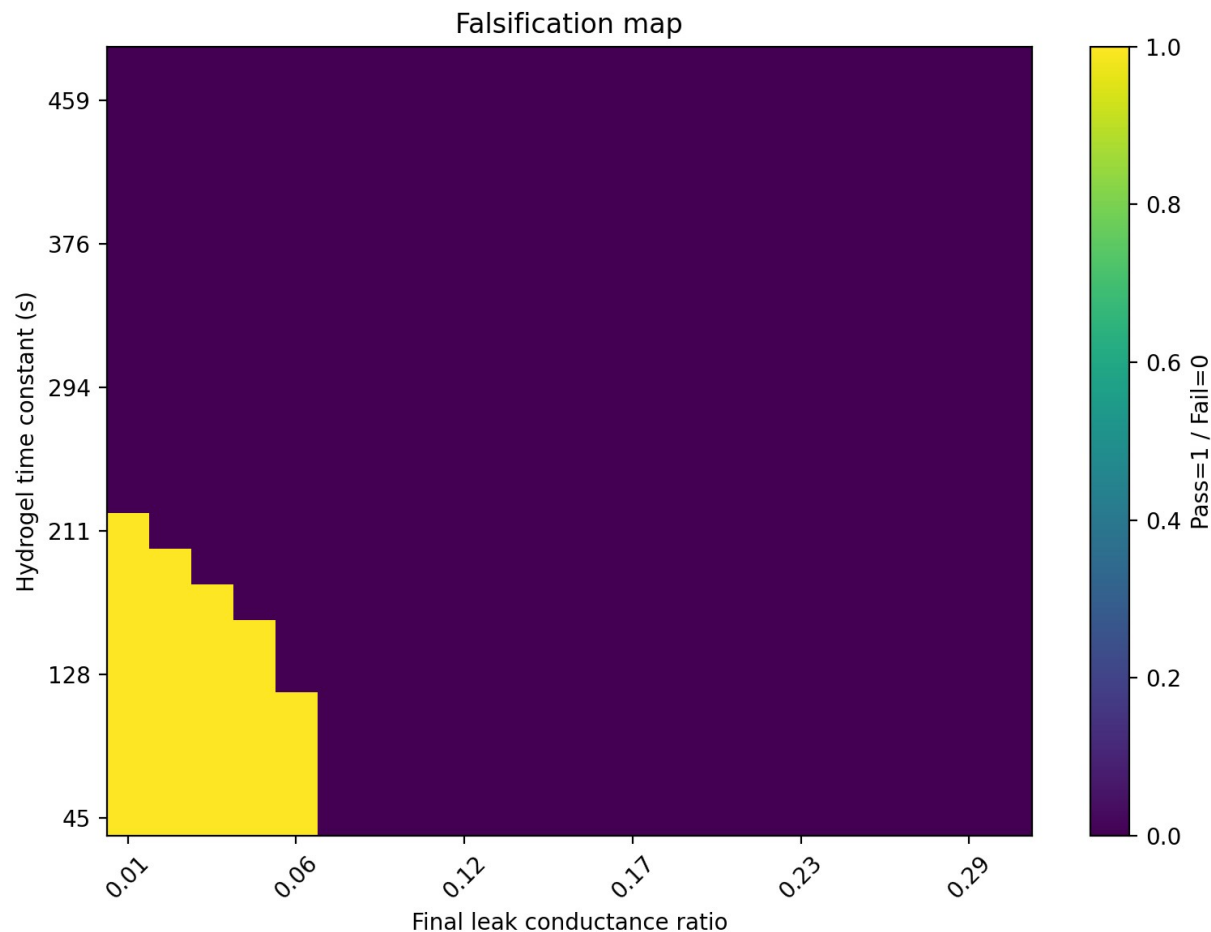


Figure S1. Parameter-region pass/fail map for hydraulic criteria.

The supplied grid contains 484 deterministic parameter combinations; 34 satisfy the hydraulic gate under the fixed background network realization used for this map. The full grid is provided in data/falsification_parameter_grid.csv so that boundary choices can be recomputed or criticized.

S4. Proposed prototype geometry

Table S2. V1 geometry targets.

Feature	Initial target	Reason for target	Must be measured
Panel footprint	100 x 100 mm	Easy optical access and benchtop climate testing	Actual flatness and seal integrity
Vascular channel diameter-equivalent	180-250 μ m	Near model scale while more fabricable than very small channels	Hydraulic resistance and priming pressure
Branch pitch	10-20 mm	Visible routing and space for collar cartridges	Manufacturing tolerance
Damage orifice	Interchangeable/calibrated, not arbitrary puncture	Enables comparable conductance across trials	Flow-pressure calibration curve

Feature	Initial target	Reason for target	Must be measured
Collar cavity	Interchangeable annular chamber around damage branch	Allows material screening without rebuilding network	Free vs confined swelling and closure force
Bypasses	At least two alternate routes around damage node	Tests graceful degradation	Post-damage route flow distribution
Top collector coupon	Removable	Prevents fog/cooling tests from confounding core hydraulics	Capture efficiency and drainage losses

S5. Bill of materials - category level

Table S3. Material categories. Exact suppliers and chemistries should be selected only after compatibility tests.

Category	Example candidate	Why it is a candidate	Do not assume
Transparent microfluidic substrate	PMMA, COC, glass, or PDMS hybrid	Optical access and accessible prototyping	Chemical compatibility, surface stability or exact pressure rating
Hydrophilic treatment	Plasma treatment, hydrophilic coating, cellulose layer	Promote capillary priming	Long-term contact angle or antifouling
Hygromorphic element	Cellulose-filled bilayer or paper/polymer laminate	Well-established humidity-responsive strategy	Exact RH threshold or lifetime
Swelling collar	PVA/polyacrylate/superabsorbent hydrogel family	Large water-driven volume change	Required response speed under confinement
Electrokinetic insert	Charged porous membrane or oxide-coated microchannel region	Pressure/flow-sensitive electrical interface	Useful power or stable long-term voltage
Electrodes	Ag/AgCl for measurement or inert electrodes depending protocol	Controlled potential measurement	That electrode chemistry is negligible
Environmental sensing	Calibrated RH/T probe; airspeed and fog characterization	Normalize auxiliary tests	That room conditions are adequate
Flow measurement	Gravimetric collection, microbalance, flow sensors	Direct mass balance	That visual dye alone quantifies performance

S6. Physical preregistration - core hydraulic experiment

Primary question: does a damage-triggered swelling collar reduce a calibrated leak while preserving network service without powered actuation?

Table S4. Core physical preregistration.

Field	Preregistered value / rule
Primary endpoint	Leak reduction at 600 s relative to early post-damage peak
Co-primary engineering endpoint	Useful outlet flow at 600 s relative to undamaged baseline
Primary success threshold	Median leak reduction $\geq 80\%$ and median useful-flow retention $\geq 70\%$, with individual traces reported
Timing endpoint	Time to first sustained $\geq 80\%$ leak reduction
Control 1	Identical damage geometry without swelling collar
Control 2	Sham puncture / handling without opening leak path
Randomization	Randomize specimen test order and collar/control order where feasible
Blinding	Automate trace extraction; analyze coded specimen IDs if feasible
Exclusions	Only pre-test fabrication/seal failures defined before endpoint inspection; report exclusions separately
Raw data	Time, inlet/outlet/leak mass or flow, temperature, RH, pressure head, video ID, specimen ID, material lot

S7. Falsification matrix

Table S5. Alternative explanations and rejection tests.

Potential explanation	Discriminating test	Outcome that falsifies desired mechanism
Leak reduction is just reservoir depletion	Maintain controlled upstream head; compare no-collar control	Leak falls equally without collar
Flow retention is an artifact of reduced total inflow	Measure inlet and outlet simultaneously	Outlet preservation disappears after mass balance
Collar swelling blocks entire network, not leak	Image/local pressure measurement around bypasses	Useful flow collapses despite leak reduction
Electrical signal is electrode motion/polarization	Sham puncture, inert insert, polarity reversal, conductivity series	Same signal occurs without hydraulic perturbation or changes inconsistently with pressure
Cooling is only initial wetting transient	Longer steady-state test with continuous mass-loss measurement	Temperature advantage vanishes once transient is excluded
Fog capture is condensation on apparatus	Blank/control collector and chamber-loss accounting	Patterned ACS does not exceed blank after normalization
Humidity gate simply softens without functional aperture	Direct image-based aperture measurement	No reproducible geometry/permeability change

S8. Raw-file manifest

- nominal_timeseries.csv: 901-row virtual time series.
- monte_carlo_unoptimized.csv: 1,000 broad-prior virtual trials.
- monte_carlo_design_envelope.csv: 1,000 refined virtual trials.
- falsification_parameter_grid.csv: 484 deterministic parameter combinations.
- hygromorphic_gate_prediction.csv: Parametric RH response.
- evaporative_cooling_prediction.csv: Cooling bounds.
- fog_collection_bounding_predictions.csv: Fog-capture bounds.

S9. Reporting checklist for the first physical paper

- ☐ State clearly that V0.1 was computational and predated the prototype
- ☐ Publish every preregistered endpoint even if negative
- ☐ Upload unprocessed raw traces and calibration files
- ☐ Report exact hydrogel chemistry, dimensions, conditioning, temperature and water chemistry
- ☐ Report pressure/flow calibration of each damage orifice
- ☐ Provide source impedance and load curve for any electrical-output claim
- ☐ Separate “self-sensing” from “energy harvesting” unless usable electrical power is measured
- ☐ Report water mass balance for any cooling claim
- ☐ Report environmental chamber conditions for atmospheric-water claims
- ☐ Distinguish model calibration specimens from independent prediction specimens

S10. Computational result summary

Table S6. Key numerical outputs. All are computational.

Metric	Virtual result
Nominal leak reduction (%)	92.79
Nominal useful-flow retention (%)	97.38
Nominal 80% isolation time (s)	276
Nominal virtual sensor SNR	235.1
Broad-prior pass rate (%)	27.3
Design-envelope pass rate (%)	99.7

S11. Safety and scope

The first prototype should use water and non-hazardous dyes only. Biological aerosols, pathogens, toxic pesticides or environmental release are unnecessary for validating the central ACS hypothesis and should not be introduced into early testing. Later application-specific testing would require the relevant biosafety, chemical-safety and environmental procedures.

AI assistance and audit trail. The package was produced with AI assistance for numerical implementation, figure generation and drafting. Explicit source, random seeds, raw outputs and falsification gates are included for auditability. Future physical work should add a laboratory notebook, calibration records, specimen provenance and version-controlled analysis scripts.